7. THE TOOLS OF NUCLEAR PHYSICS

I. THE METHODS OF DETECTION AND OBSERVATION

The following sections of this book will deal with the tools and methods available to the nuclear physicist both for generating and observing the phenomena discussed in the preceding lectures. These procedures call for immense quantities of energy, and the most powerful instruments known to technical science must be used to supply them. Yet, the material results achieved even with these vast stores of energy are extremely small. Consequently, for these studies it is indispensable to have extraordinarily sensitive instruments, for the phenomena which are to be studied take place in one individual atom or, at best, in a very few atoms—in structures which are inconceivably small according to ordinary conceptions.

We shall begin with the instruments of detection and study. The oldest method is the *scintillation method.* When a very fast particle—an alpha particle for instance—impinges on a zinc sulphide screen, a reaction occurs there which produces a weak flash of lightning, a *scintillation.* It is therefore possible to observe the impact of each individual particle of sub-atomic order of magnitude, as the impact of bullets on a plastered wall can be observed, and we can also count the particles this way. It is, however, a poor policy to depend on the unaided eye, which is bound to grow tired little by little during the process of counting. This method is scarcely ever employed these days. It has been taken up recently by recording the weak flashes not with the eye but by electric amplification.

The *ionization chamber* supplies the fundamental principle on which most of the modern methods of observation are based. Let us attempt to describe this apparatus in a very rudimentary form: A gold-leaf electroscope consists of an earthed metal box containing an insulated metal rod with two gold leaves which spread apart when an electric charge reaches the rod. (Figure 29.) An inverted metal hood, charged by a battery to about 100 volts relatively to the earth, is placed above the electroscope and insulated from the latter. When a charged particle—an alpha or beta particle, or even a gamma-ray photon—enters the space between the electroscope and the hood, it tears electrons off the air molecules. These electrons attach themselves to other molecules and form negative ions, while the molecules thus deprived of electrons remain behind as positive ions. As shown in Figure 29, such ions are created all along the track of the particle. If the hood is positively charged, the positive ions stream to the rod of the electroscope, and the negative ions to the